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12.4. Biomedical Techniques

Blood Cell Counting using Haemocytometer

The haemocytometer is a thick glass slide with a counting chamber in the middle. The counting chamber contains two grids with improved Neubaur rulings of 3 by 3 mm primary square. The primary square is further subdivide into 9 secondary squares, each 1 by 1 mm. The four corner squares are used for the white blood cell count which are further subdivided into 16 tertiary squares. The central secondary square is divided into 25 tertiary squares, each of which measure 0.2 by 0.2 mm, each single tertiary square is further divided into 16 smaller squares. The five black squares along with the shaded squares in the centre are used for platelet count, while the five black squares alone are used for red blood cells counting (Figure12.10).

Diluting fluid

The blood cells are diluted in specific diluting fluid to keep the cells intact. RBC diluting fluid (Hayem's) is isotonic with blood, hence haemolysis does not occur. The blood is diluted 1:200 times with RBC diluting fluid and the cells are counted under 45X objective of the microscope.

The diluting fluid used for WBC count is Turk's solution which contains glacial acetic acid and Gentian violet. The glacial acetic acid lyses the red blood cells and the Gentian violet stains the nuclei of the leucocytes. The blood is diluted 1:20 times and the cell are counted under 10X objective of the microscope. The total number of cells counted is expressed in mm^3.

Laser in-situ keratomileusis (LASIK) is a non-reversible refractive procedure performed by ophthalmologists to correct myopia, hypermetropia and astigmatism. The surgeon uses an excimer laser to cut or reshape the cornea so that light will focus properly on the retina. Many professional athletes have chosen LASIK to improve their performance.

Blood cell counting using hemocytometer

1. The blood is collected till the 0.5 graduation in the pipette.
2. The diluting fluid is taken till the graduations 11 and 101 of the WBC and RBC pipette respectively.
3. The blood is diluted and mixed well with the respective diluting fluid by rotating the pipette horizontally several times.
4. The cover slip is placed on top of the counting chamber.
5. The tip of the pipette is placed on the counting chamber and fluid discharged till it fills the chambers.
6. The cells are allowed to settle for several minutes and the ruled area is viewed under the microscope.

A person suffering from severe fever, head ach and fatigue so he visits his doctor. What type of diagnosing procedure does the doctor recommended for him. Name the medical equipment is used for it?